

CI/CD Observability in Containerized Environments

Eduardo Sousa

**A dissertation preparation for the degree of
of Master of Science, Specialisation Area of Cybersecurity And
Systems Administration**

**Advisor: Dr. Alexandre Bragança
Supervisor: Dr. Joao Silva**

Statement of Integrity

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I have not plagiarised or applied any form of undue use of information or falsification of results along the process leading to its elaboration.

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ISEP, Porto, December 31, 2025

Abstract

This document explains the main formatting rules to apply to a TMDEI Master Dissertation work for the MSc in Computer Engineering of the Computer Engineering Department (DEI) of the School of Engineering (ISEP) of the Polytechnic of Porto (IPP).

The rules here presented are a set of recommended good practices for formatting the dissertation work. Please note that this document does not have definite hard rules, and the discussion of these and other aspects of the development of the work should be discussed with the respective supervisor(s).

This document is based on a previous document prepared by Dr. Fátima Rodrigues (DEI/ISEP).

The abstract should usually not exceed 200 words, or one page. When the work is written in Portuguese, it should have an abstract in English.

Please define up to 6 keywords that better describe your work, in the *THESIS INFORMATION* block of the `main.tex` file.

Keywords: CI/CD, Observability, Containers, Tracing, Logging, Monitoring

Resumo

Trabalhos escritos em língua Inglesa devem incluir um resumo alargado com cerca de 1000 palavras, ou duas páginas.

Se o trabalho estiver escrito em Português, este resumo deveria ser em língua Inglesa, com cerca de 200 palavras, ou uma página.

Para alterar a língua basta ir às configurações do documento no ficheiro `main.tex` e alterar para a língua desejada ('english' ou 'portuguese')¹. Isto fará com que os cabeçalhos incluídos no template sejam traduzidos para a respetiva língua.

¹Alterar a língua requer apagar alguns ficheiros temporários; O target **clean** do **Makefile** incluído pode ser utilizado para este propósito.

Acknowledgement

The optional Acknowledgment goes here. . . Below is an example of a humorous acknowledgment.

"I'd also like to thank the Van Allen belts for protecting us from the harmful solar wind, and the earth for being just the right distance from the sun for being conducive to life, and for the ability for water atoms to clump so efficiently, for pretty much the same reason. Finally, I'd like to thank every single one of my forebears for surviving long enough in this hostile world to procreate. Without any one of you, this book would not have been possible." in "The Woman Who Died a Lot" by Jasper Fforde.

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a	distance	m
P	power	W (Js^{-1})
ω	angular frequency	rad

Chapter 1

Thesis Structure

1.1 Introduction

Internal Continuous Integration / Continuous Delivery (CI/CD) services (GitLab, Jenkins, SonarQube, Nexus, Dependency-Track) deployed in Docker containers are critical to software delivery pipelines. Reliability issues directly affect development velocity and release cadence. However, constrained deployment contexts, non-privileged containers, no host administrative access, and heterogeneous operating systems (Linux and Windows).

This research examines integrated observability (systematic collection, correlation and analysis of metrics, logs and distributed traces) in such constrained environments. The proposed solution combines non-privileged telemetry collection (sidecar exporters, agent collectors, runtime APIs) with correlation mechanisms to improve Mean Time To Detection (MTTD), Mean Time To Recovery (MTTR), and pipeline throughput while maintaining sub-5% resource overhead. The work is grounded in a real-world, four-server on-premise deployment (three Linux, one Windows) with no administrative access, exemplifying operational constraints typical of internally-managed infrastructure where third-party services deployed via Docker Merkel 2014 cannot be modified.

The central research problem is: *How can an integrated observability framework be designed and implemented for containerized CI/CD platforms operating under non-privileged access constraints and heterogeneous infrastructure, to measurably improve reliability and reduce incident response times?*

1.2 Problem Statement

1.3 Research Objectives

1.4 Research Question

1.5 Methodology Overview

1.6 Ethical Considerations

1.7 Structure of the Dissertation

Chapter 2

Skill Management and Project Planning

2.1 Skill Management

2.1.1 Skills Identification

2.1.2 Skills Assessment

2.1.3 Strategy to Improve Skills

2.2 Project Management

2.2.1 Project Charter

2.2.2 Scope of Work: WBS

2.2.3 Integrated Project Schedule

2.2.4 Monitoring and Controlling Procedures

2.2.5 Risk Identification and Management

Chapter 3

Literature review

3.1 PRISMA Methodology Overview

3.1.1 Identification

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3.2 Prisma Flow Diagram

3.3 Related Work on Observability in CICD Environments

3.3.1 Observability Frameworks

3.3.2 Telemetry Collection Techniques

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3.4 Gaps in Existing Research

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Chapter 4

Conclusion

Bibliography

Merkel, Dirk (Mar. 2014). "Docker: lightweight Linux containers for consistent development and deployment". In: *Linux J.* 2014.239. ISSN: 1075-3583.

Appendix A

Appendix Title Here

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